# Waveform tomography

# From last time

Using Fermat's principle, we find that:

$$\delta T = -\int_{\text{ray}} \frac{\delta v}{v^2}\,\mathrm{d}s,$$

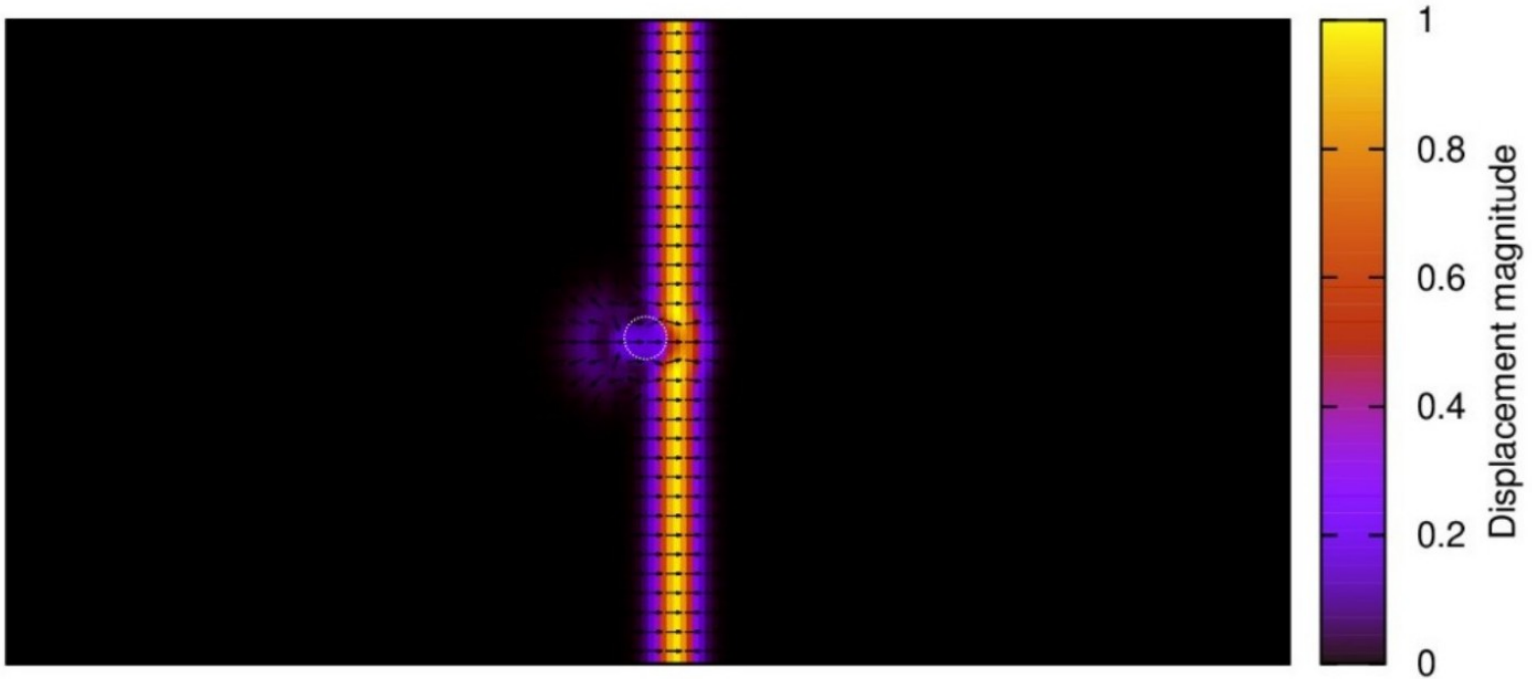
Expand the velocity in a finite-dimensional basis:

$$\delta\alpha(\mathbf{x}) = \sum_{j=1}^{m} \delta\alpha_j \varphi_j(\mathbf{x}).$$

Formulation as a regularised least squares problem:

$$J(\mathbf{m}) = \frac{1}{2}(\mathbf{A}\mathbf{m} - \mathbf{d})^T \mathbf{C}^{-1}(\mathbf{A}\mathbf{m} - \mathbf{d}) + \frac{\lambda}{2}(\mathbf{m} - \mathbf{m}_0)^T \mathbf{B}(\mathbf{m} - \mathbf{m}_0),$$

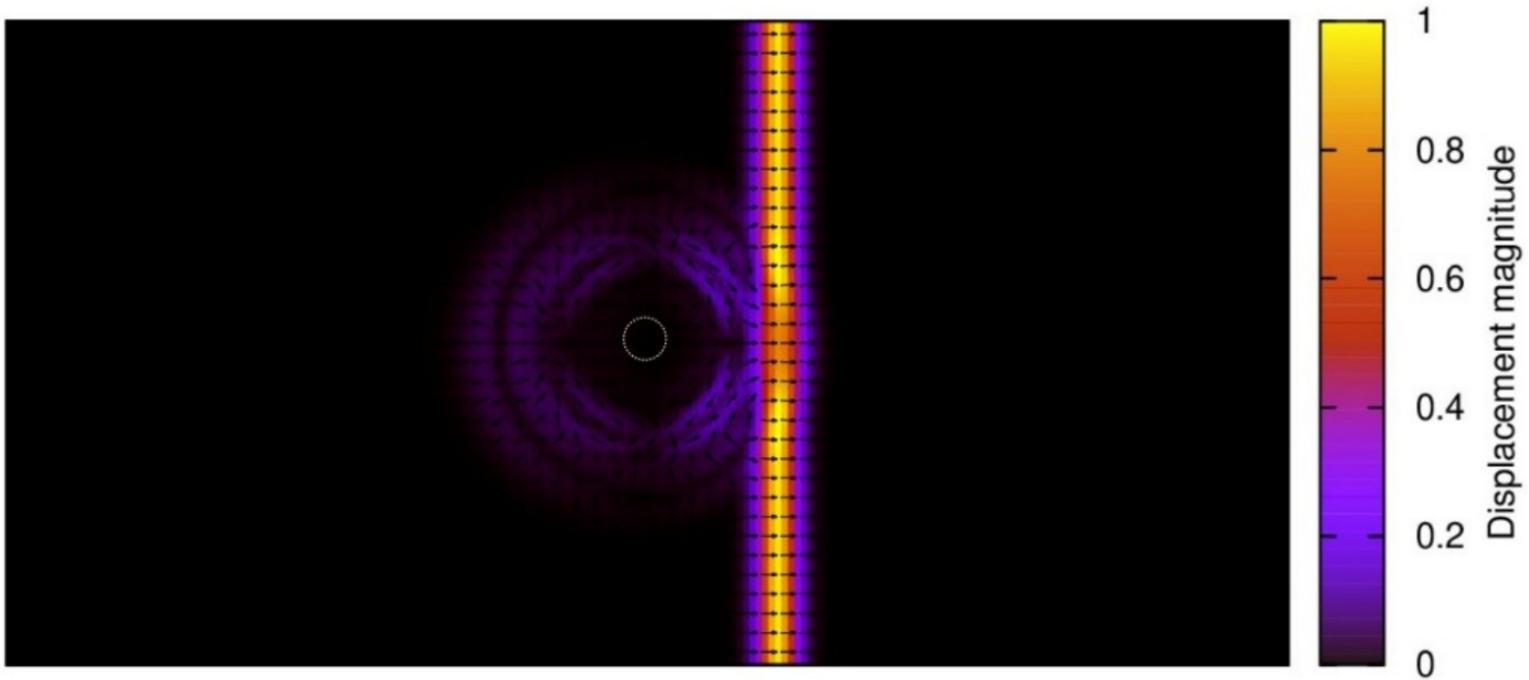
# Wavefront healing




Scattering regime: $\frac{L}{\lambda} \approx 1$

Geometric optics regime: $\frac{L}{\lambda} \gg 1$

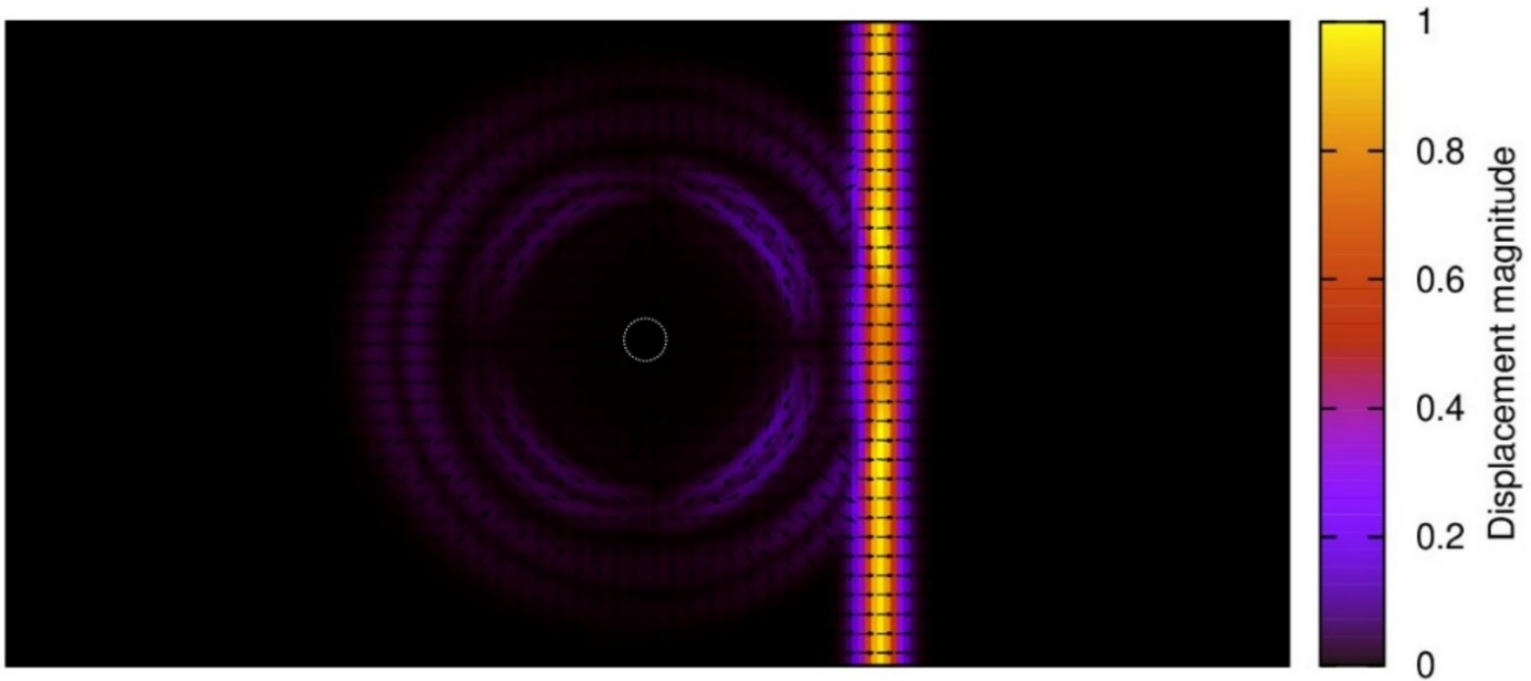
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Scattering regime: $\frac{L}{\lambda} \approx 1$

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# Wavefront healing



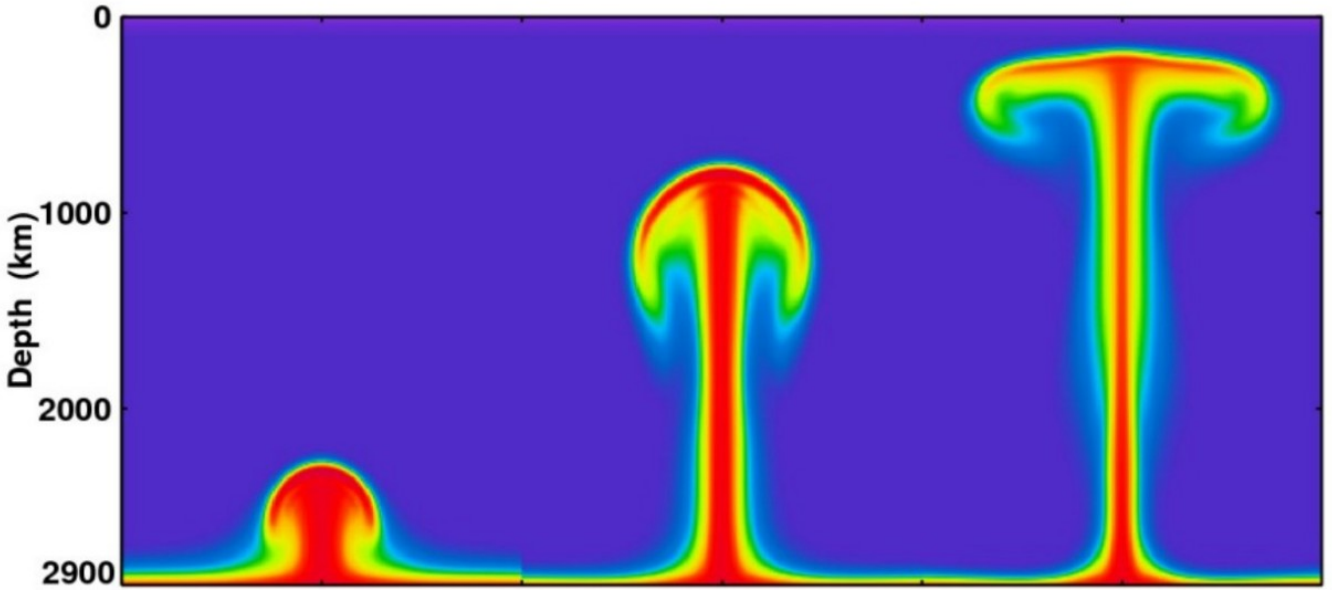

Scattering regime: $\frac{L}{\lambda} \approx 1$

Geometric optics regime: $\frac{L}{\lambda} \gg 1$

# Lengths scales in the mantle

Typical periods for global seismology: 1-40s

Corresponding wavelengths: 10-400km




Farnetani 1997

# Waveform inversion

# Equations of motion

Equations of motion:

$$\rho \frac{\partial^2 u_i}{\partial t^2} - \frac{\partial}{\partial x_j}\left( A_{ijkl} \frac{\partial u_k}{\partial x_l} \right) = -\frac{\partial \overline{S}_{ij}}{\partial x_j},$$

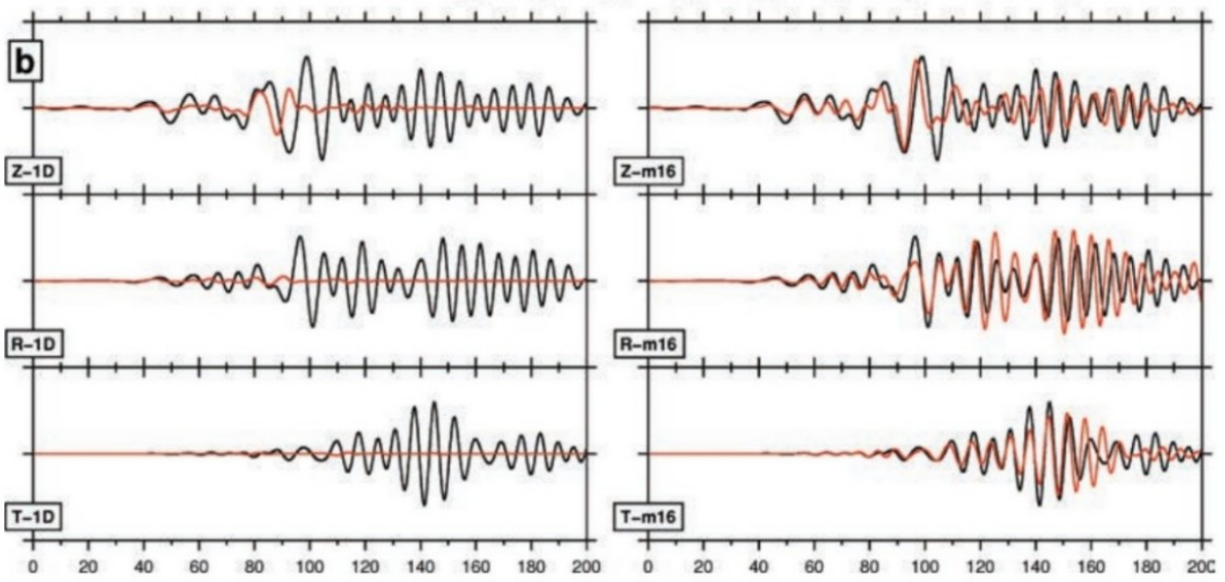
$$A_{ijkl} \frac{\partial u_k}{\partial x_l} \hat{n}_j = \overline{S}_{ij} \hat{n}_j$$

These can be solved numerically in realistic laterally heterogeneous earth models using techniques like the **finite element methods**

These are demanding calculations, run on HPC systems using parallel computing over hundreds or thousands of cores. Applications of GPU computing is common

Each simulation take **tens of minutes to hours** depending on resources

# Waveform misfits




Tape et al. 2010

$$J = \frac{1}{2}\int_0^T ||\mathbf{u}(\mathbf{x}_r, t) - \mathbf{u}^{obs}(t)||^2 dt,$$

# Formulation of the problem

Misfit is an implicit function of the parameters in the elastic wave equation:

1) **Density** – piecewise constant function
2) **Elastic tensor** – piecewise constant tensor field (21 independent components)
3) **Stress glut** – space and time dependent tensor field
4) **Geometry** of the model (e.g., topography on internal boundaries)

Relation can be condensed to:

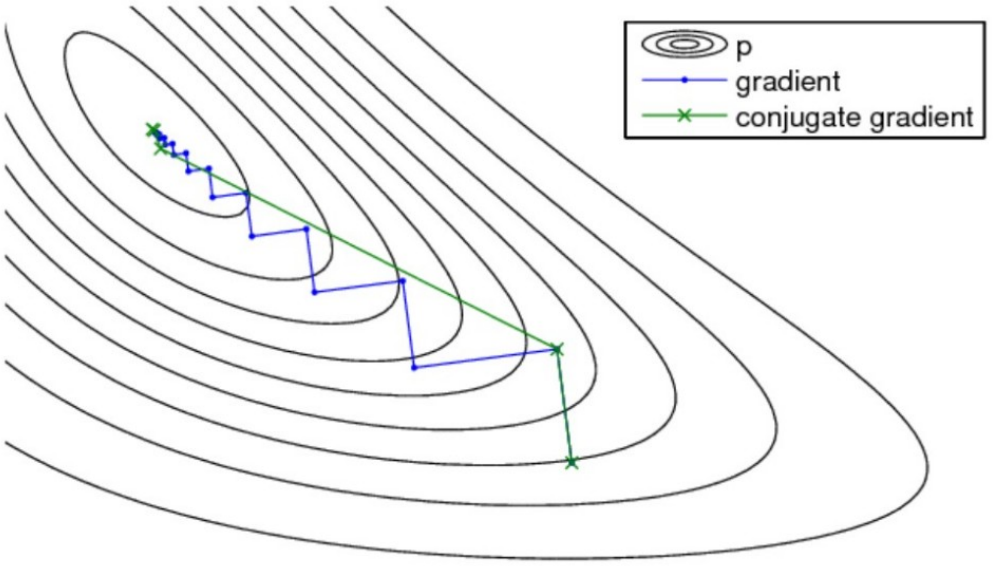
$$\mathbf{m} \mapsto J(\mathbf{m})$$

The aim is to **minimise** this functional

Strictly defined on an **infinite-dimensional space**. In practice reduced to a finite but high-dimensional space linked to the numerical discretisation of the equations of motion

**Millions or billions of degrees of freedom would be typical**

# Gradient based optimisation




Iterative reduction of the misfit using the local gradient to take steps downhill

For large-scale optimisation problems, this is often the only viable approach

Bayesian sampling methods are not typically an option, though this is activate research topic

To proceed, we need a way of evaluating the gradients

# Functional derivatives

Functional derivative defined through a first-order Taylor expansion:

$$J(\mathbf{m}+\delta\mathbf{m}) = J(\mathbf{m}) + \langle DJ(\mathbf{m})|\delta\mathbf{m}\rangle + O(||\delta\mathbf{m}||^2),$$

The terms **gradient** and **sensitivity kernel** are equivalent and widely used

Focusing on S-wave velocity variations alone, we can write to first order:

$$\langle DJ(\mathbf{m})|\delta\mathbf{m}\rangle = \int_M K_\beta \delta\beta \, \mathrm{d}^3\mathbf{x}$$

Methods for calculation will be discussed later

# Gradient based optimisation

Condition for a minimum – global and then local:

$$J(\overline{\mathbf{m}} + \delta\mathbf{m}) \geq J(\overline{\mathbf{m}}),$$

$$DJ(\overline{\mathbf{m}}) = \mathbf{0}.$$

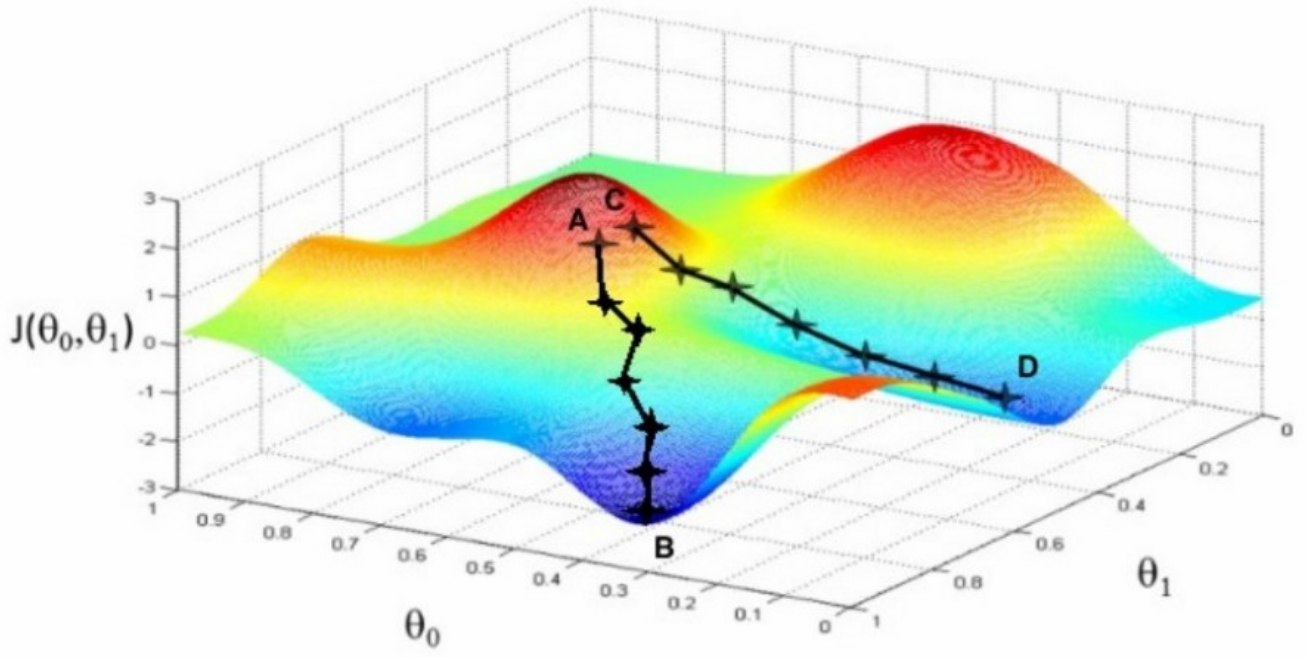
Starting an an initial point, we consider the update:

$$\mathbf{m}_1 = \mathbf{m}_0 - \lambda\, DJ(\mathbf{m}_0)$$

Step length determined by approximately solving a 1D minimisation problem

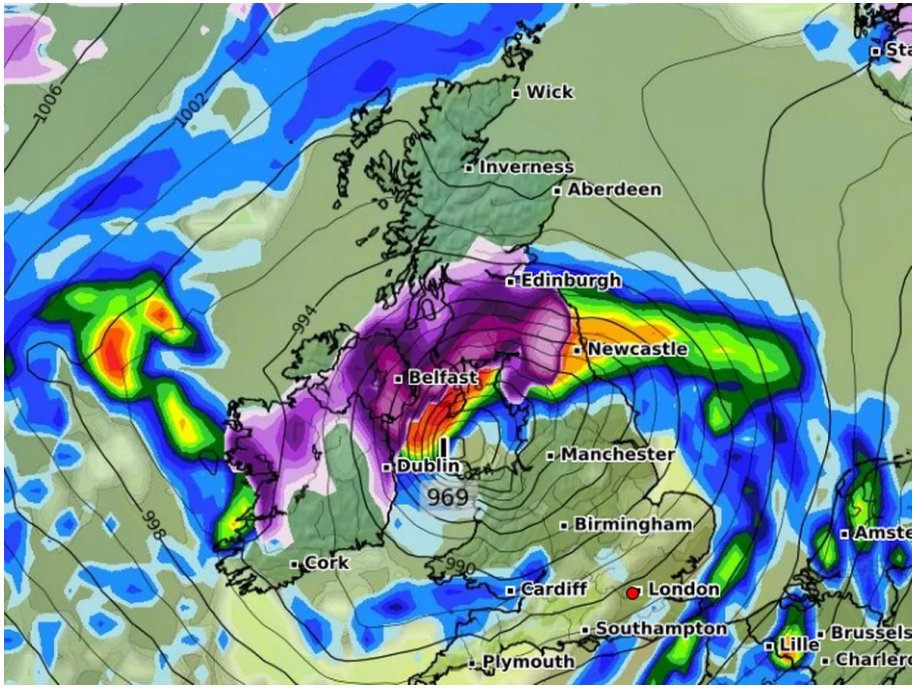
Process is repeated until convergence or the misfit is sufficiently small

# Local minima




Gradient based methods produce only local minima

Uncertainty quantification possible to a limit extent, but adds significantly to the cost







Essentially the same methods underlie weather forecasting along with a host of other important applications

# Calculating the gradients

# How not to do it

Finite difference approximation:

$$\langle DJ(\mathbf{m})|\delta\mathbf{m}\rangle \approx J(\mathbf{m}+\delta\mathbf{m}) - J(\mathbf{m}).$$

Approximation of derivative relative to an **n-dimensional model parametrisation** requires **n+1 forward calculations**

One gradient calculation using this method for a realistic problem would take of order hundreds of years on a massive HPC system!

The **adjoint method** provides the exact gradient at a cost of **two forward simulations**

# PDE constrained optimisation

We want to minimise:

$$J = \frac{1}{2}\int_0^T ||\mathbf{u}(\mathbf{x}_r, t) - \mathbf{u}^{obs}(t)||^2 dt,$$

subject to the constraints:

$$\rho\frac{\partial^2 u_i}{\partial t^2} - \frac{\partial}{\partial x_j}\left(A_{ijkl}\frac{\partial u_k}{\partial x_l}\right) = -\frac{\partial \overline{S}_{ij}}{\partial x_j},$$

$$A_{ijkl}\frac{\partial u_k}{\partial x_l}\hat{n}_j = \overline{S}_{ij}\hat{n}_j$$

This can be done using the **method of Lagrange multipliers**

# PDE constrained optimisation

We start by defining the **Lagrangian**:

$$L(\mathbf{u}, \mathbf{u}', \mathbf{w}', \mathbf{m}) = J(\mathbf{u}) - \int_0^T \int_M \left[ \rho \frac{\partial^2 u_i}{\partial t^2} - \frac{\partial}{\partial x_j} \left( A_{ijkl} \frac{\partial u_k}{\partial x_l} - \overline{S}_{ij} \right) \right] u_i' \, \mathrm{d}^3\mathbf{x} \, \mathrm{d}t$$
$$- \int_0^T \int_{\partial M} \left( A_{ijkl} \frac{\partial u_k}{\partial x_l} - \overline{S}_{ij} \right) \hat{n}_j w_i' \, \mathrm{d}S \, \mathrm{d}t.$$

Each constraint is associated with a **Lagrange multiplier field**

Zero initial conditions on displacement and velocity are being imposed directly

Within the Lagrangian, all fields are regarded as being independent

# Lagrange multiplier theorem

The following equality holds:

$$DJ(\mathbf{m}) = D_{\mathbf{m}} L(\mathbf{u}, \mathbf{u}', \mathbf{w}', \mathbf{m}),$$

so long as the following conditions are met:

$$\begin{aligned} D_{\mathbf{u}} L(\mathbf{u}, \mathbf{u}', \mathbf{w}', \mathbf{m}) &= 0, \\ D_{\mathbf{u}'} L(\mathbf{u}, \mathbf{u}', \mathbf{w}', \mathbf{m}) &= 0, \\ D_{\mathbf{w}'} L(\mathbf{u}, \mathbf{u}', \mathbf{w}', \mathbf{m}) &= 0. \end{aligned}$$

Subscripts used for partial functional derivatives

You only need to know how to apply this method **not to prove** that it works – details provided in an appendix to the handout for anyone interested

# The forward problem

Recall that the Lagrangian is:

$$L(\mathbf{u},\mathbf{u}',\mathbf{w}',\mathbf{m}) = J(\mathbf{u}) - \int_0^T \int_M \left[\rho\frac{\partial^2 u_i}{\partial t^2} - \frac{\partial}{\partial x_j}\left(A_{ijkl}\frac{\partial u_k}{\partial x_l} - \overline{S}_{ij}\right)\right] u_i' \,\mathrm{d}^3\mathbf{x}\,\mathrm{d}t$$
$$- \int_0^T \int_{\partial M} \left(A_{ijkl}\frac{\partial u_k}{\partial x_l} - \overline{S}_{ij}\right) \hat{n}_j w_i' \,\mathrm{d}S\,\mathrm{d}t.$$

Stationarity condition wrt to **u**' gives the equations of motion

Stationary conditoins wrt to **w**' give the boundary conditions

This is how the forward problem emerges within the method

# The adjoint problem

Varying wrt to **u** gives:

$$\langle D_{\mathbf{u}} L(\mathbf{u}, \mathbf{u}', \mathbf{w}', \mathbf{m}) | \delta \mathbf{u} \rangle = \langle DJ(\mathbf{u}) | \delta \mathbf{u} \rangle - \int_0^T \int_M \left[ \rho \frac{\partial^2 \delta u_i}{\partial t^2} - \frac{\partial}{\partial x_j} \left( A_{ijkl} \frac{\partial \delta u_k}{\partial x_l} \right) \right] u_i' \, \mathrm{d}^3\mathbf{x} dt$$
$$- \int_0^T \int_{\partial M} \left( A_{ijkl} \frac{\partial \delta u_k}{\partial x_l} \right) \hat{n}_j w_i' \, \mathrm{d}S \, \mathrm{d}t,$$

From the chosen form for *J* we can write:

$$\langle DJ(\mathbf{u}) | \delta \mathbf{u} \rangle = \int_0^T \int_{\partial M} h_i' \delta u_i \, \mathrm{d}S \, \mathrm{d}t,$$

$$h_i'(\mathbf{x}, t) = [u_i(\mathbf{x}_r, t) - u_i^{obs}(t)] \delta(\mathbf{x} - \mathbf{x}_r).$$

# The adjoint problem

We need the following condition to hold:

$$\int_0^T \int_{\partial M} h_i' \delta u_i \, \mathrm{d}S \, \mathrm{d}t - \int_0^T \int_M \left[ \rho \frac{\partial^2 \delta u_i}{\partial t^2} - \frac{\partial}{\partial x_j} \left( A_{ijkl} \frac{\partial \delta u_k}{\partial x_l} \right) \right] u_i' \, \mathrm{d}^3\mathbf{x} \, \mathrm{d}t$$
$$- \int_0^T \int_{\partial M} \left( A_{ijkl} \frac{\partial \delta u_k}{\partial x_l} \right) \hat{n}_j w_i' \, \mathrm{d}S \, \mathrm{d}t = 0,$$

For all variations of the displacement vector

To see the consequence, we need to pass all the derivatives to the other terms using integration by parts in time and the divergence theorem spatially

Full details are in the handout

# The adjoint problem

The result of the long calculation is that:

$$
\begin{aligned}
&-\int_0^T \int_M \left[ \rho \frac{\partial^2 u_i'}{\partial t^2} - \frac{\partial}{\partial x_j} \left( A_{ijkl} \frac{\partial u_k'}{\partial x_l} \right) \right] \delta u_i \, \mathrm{d}^3\mathbf{x} \, \mathrm{d}t \\
&-\int_0^T \int_{\partial M} \left( A_{ijkl} \frac{\partial \delta u_k}{\partial x_l} \right) \hat{n}_j (w_i' - u_i') \, \mathrm{d}S \, \mathrm{d}t \\
&+\int_0^T \int_{\partial M} \left[ h_i' - \left( A_{ijkl} \frac{\partial u_k'}{\partial x_l} \right) \hat{n}_j \right] \delta u_i \, \mathrm{d}S \, \mathrm{d}t \\
&-\int_M \rho \left\{ \left[ \frac{\partial \delta u_i}{\partial t} u_i' \right]_{t=T} - \left[ \delta u_i \frac{\partial u_i'}{\partial t} \right]_{t=T} \right\} \mathrm{d}^3\mathbf{x} = 0,
\end{aligned}
$$

From this we can read off the necessary adjoint equations

# The adjoint problem

We have shown that the Lagrange multipliers must satisfy the equations:

$$\rho \frac{\partial^2 u_i'}{\partial t^2} - \frac{\partial}{\partial x_j}\left( A_{ijkl} \frac{\partial u_k'}{\partial x_l} \right) = 0.$$

$$\left( A_{ijkl} \frac{\partial u_k'}{\partial x_l} \right) \hat{n}_j = h_i', \qquad u_i'(\mathbf{x}, T) = \frac{\partial u_i'}{\partial t}(\mathbf{x}, T) = 0.$$

along with the equality **u**' = **w**' on the boundary

This is a **terminal-value problem** for **u**' driven by the boundary traction **h**':

$$h_i'(\mathbf{x}, t) = [u_i(\mathbf{x}_r, t) - u_i^{obs}(t)]\delta(\mathbf{x} - \mathbf{x}_r).$$

# Sensitivity kernels

Using the multiplier theorem and integrating by parts, we find that:

$$\langle DJ(\mathbf{m})|\delta\mathbf{m}\rangle = \int_0^T \int_M \left[ \delta\rho \frac{\partial u_i}{\partial t}\frac{\partial u_i'}{\partial t} - \delta A_{ijkl}\frac{\partial u_i'}{\partial x_j}\frac{\partial u_k}{\partial x_l} + \delta\overline{S}_{ij}\frac{\partial u_i'}{\partial x_j} \right] \mathrm{d}^3\mathbf{x}\,\mathrm{d}t,$$

We can then read-off the sensitivity kernels:

$$K_\rho = \int_0^T \frac{\partial u_i}{\partial t}\frac{\partial u_i'}{\partial t} dt,$$

$$K_{A_{ijkl}} = -\int_0^T \frac{\partial u_i'}{\partial x_j}\frac{\partial u_k}{\partial x_l} dt,$$

$$K_{\overline{S}_{ij}} = \frac{\partial u_i'}{\partial x_j}.$$

# Summary of the adjoint method

For each gradient calculation we must:

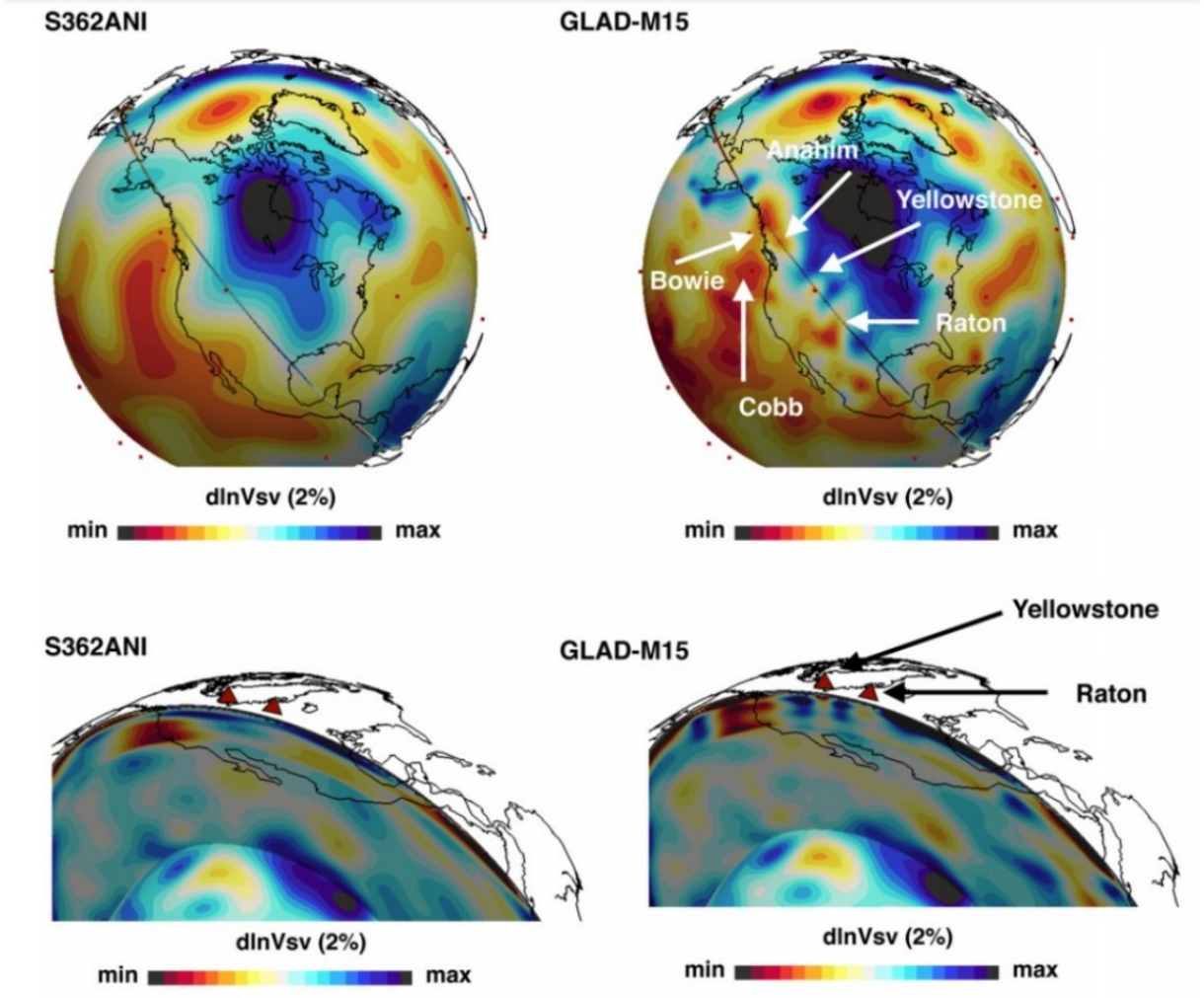
1) Solve the forward problem for **u**
2) From this solution, compute *J,* and form the adjoint traction **h**’
3) Solve the adjoint problem for **u**’
4) Combine the forward and adjoint fields to form the sensitivity kernels

Such a calculation has the cost of two forward calculations

Adjoint equation near identical to the forward one, and so the same numerical code can be used with only minor modification

Choice of misfit enters into the problem only through the form of the adjoint traction

**Calculations that would take hundred of years reduced to a minutes or hours**




Bozdag *et al.* 2016 – first global waveform inverse model

# Banana doughnut kernels

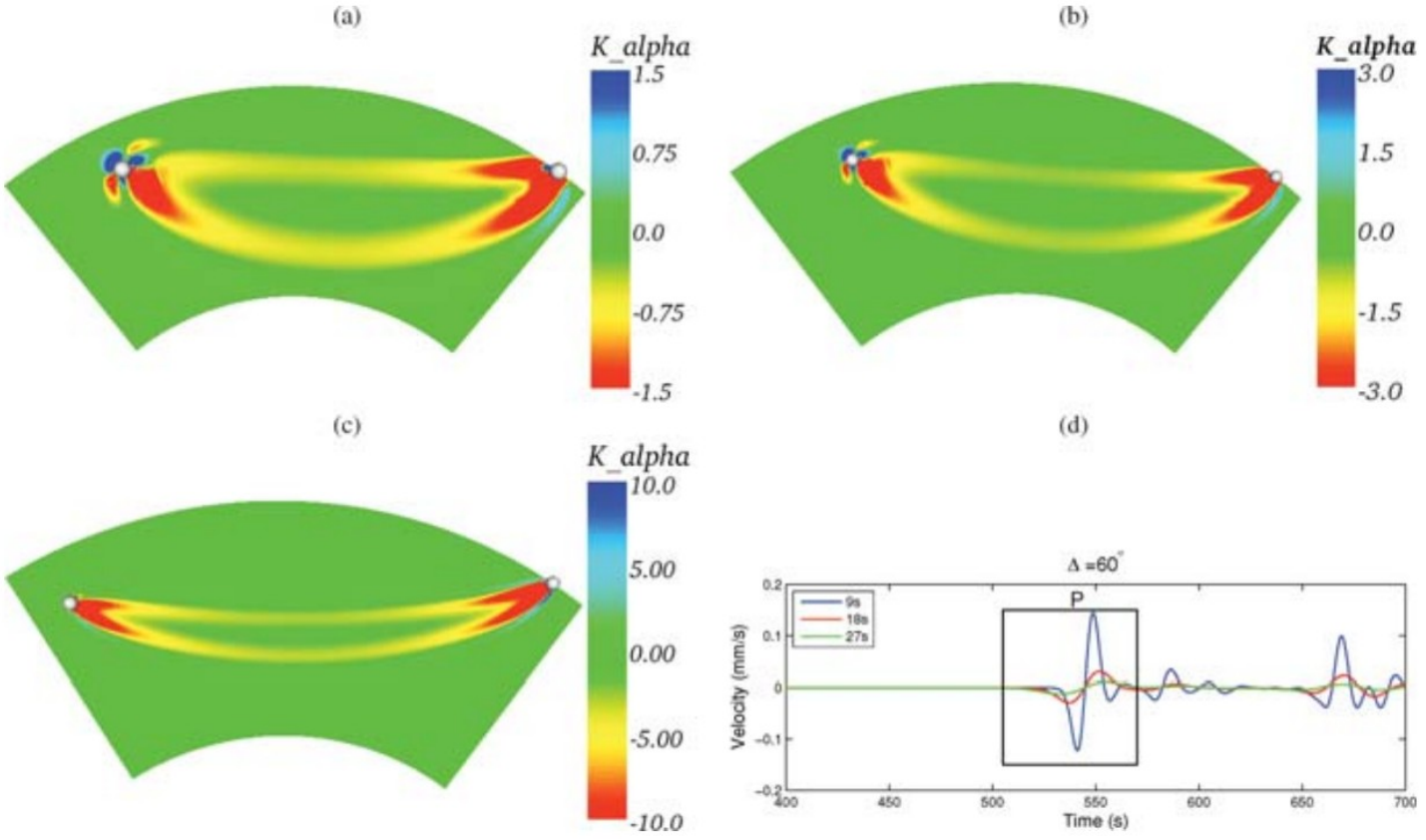
In practice, delay times are measured using **cross-correlations**:

$$C(\tau) = \int_{-\infty}^{\infty} s^{obs}(t-\tau)s(\mathbf{x}_s, t)dt.$$

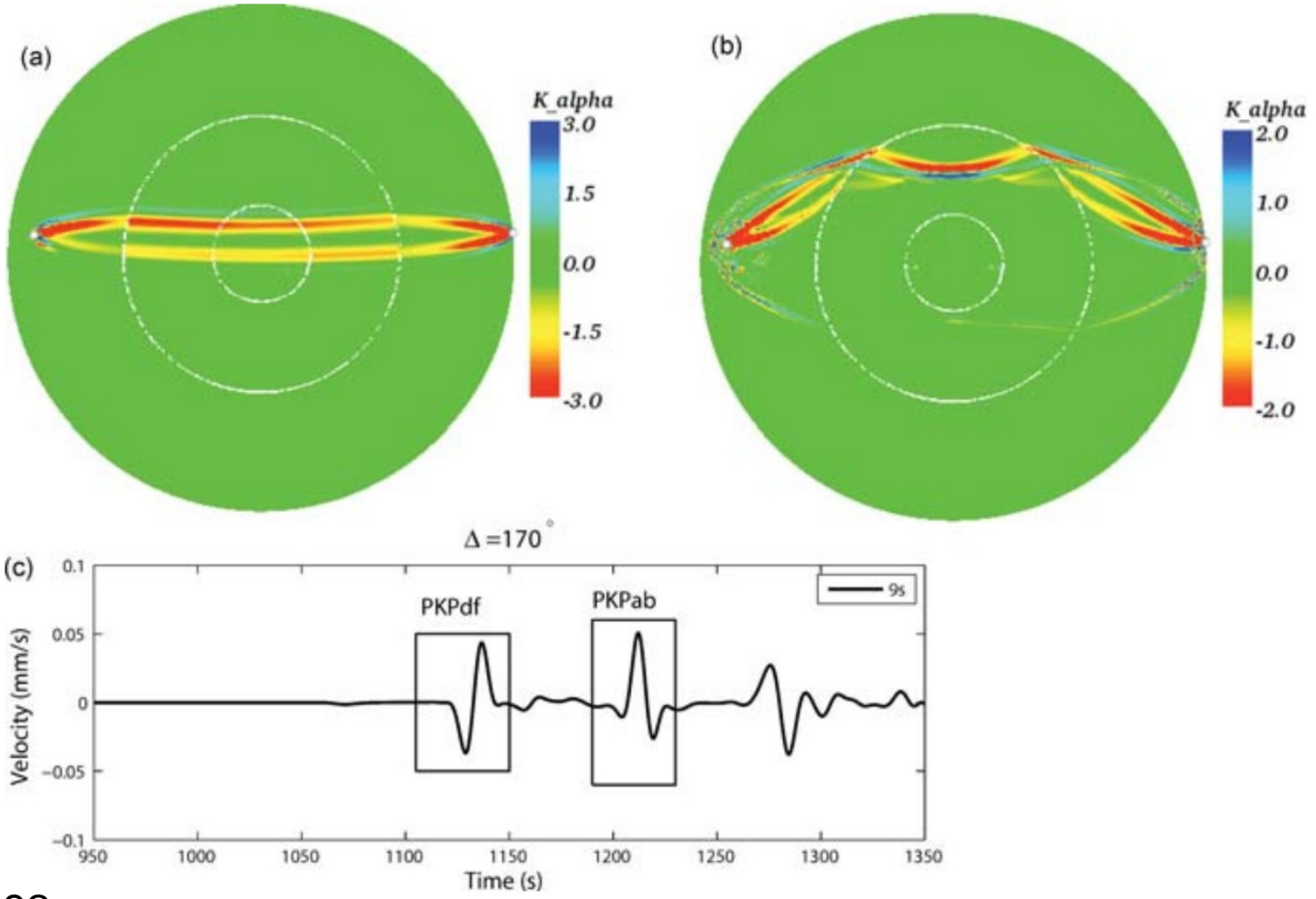
Time-shift than maximises the **cross-correlation** taken as the delay time

This defines a functional of the synthetic waveform, and hence we can derive the associated sensitivity kernels using the adjoint method

The only step we haven't covered is determining the adjoint traction for this choice of functional – this is an easy calculation, but not one you need to know




Liu & Tromp 2008




Liu & Tromp 2008